

Vaquero Marketplace Desktop App

Tech Stack:

Frontend (GUI) - Python CustomTkinter (modernized Tkinter UI)

Backend - Supabase (reused from the webapp)

Auth - Supabase

API Communication - supabase -py (Python SDK) or requests

Version Control - Github

Dependencies - customtkinter, supabase,request

Github Link

<https://github.com/francisco-020/vaquero-desktop.git>

Features - MVP Scope

1. **Core Features** - User sign up/login (Supabase Auth)
2. **Post a listing** - (title, description, price, category, contact info, image)
3. **View all listings**
4. **View all individual listing details**
5. **Filter listings by category**
6. **Filter listings by price**
7. **Sort listings by newest**
8. **Delete/edit own listings**
9. **Protect /post and /account routes**
10. **Save/bookmark a listing**
11. **Contact the seller via email or message.**
12. **Mark listing as "sold"**
13. **Filter by meet up location**

CRUD Operations

Create = Post new listings

Read = Browse/view listings

Update = Edit listing (only if user owns the listing)

Delete = Delete listing (only if user owns the listing)

Views/Windows in the Desktop App

Login/Sign up window - sign in with supabase auth or create new account

Home feed - displays all listings; supports filtering and sorting (price, category, location, newest)

Post listing - Form for creating new listings (with required fields)

Individual listing view - Shows defined info for a selected listing, with options to contact seller or bookmark

Edit listing - form pre-filled with existing data, allows owner to update or delete

Bookmarks/Favorites - Displays saved listings (only visible to the logged-in user)

Account Dashboard - View manage only the user's own listings

WireFrame

<https://www.figma.com/proto/DLNgUVqfTY3luPf7hxtcA7/Vaquero-MarketPlace?node-id=1-101&p=f&t=178G1boZn5Rd2rA1-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A101>

Listing Table

Column Name	Type	Description
id	UUID	Primary key. Default: uuid ()
user_id	UUID	Foreign Key to auth.users
title	Text	Listing title
description	Text	Full Description of item
price	Numeric	Listing price in USD
category	Text	Electronics, Books, Furniture
contact_info	Text	Email or phone number
image_url	Text	Link to Supabase Storage image
item_location	Text	"Edinburg" "Brownsville" "Both"
sold	Boolean	Defaults to false. Becomes true when sold
created_at	Timestamp	Default: now()

Bookmarks Table

Column Name	Type	Description
id	UUID	Primary Key, auto-generated
user_id	UUID	References auth.users.id
listing_id	UUID	References listings.id

created_at	Timestamp	When the bookmark was added default now()
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Auth Flow

- Email/password login using Supabase Auth
- Restrict post/account access to logged in users only
- Match listing ownership via user_id

Workflow Github branches

- main - stable production build
- Desktop dev - desktop app development
- feature/* - Feature - specific work (e.g, feature/bookmark-support)

Testing QA

- Validate signup/login/logout flows
- Test missing fields, form validation
- Verify unauthorized access is blocked (edit/delete/post)
- Confirm correct listings display for each user
- Bookmark and unbookmark logic
- Confirm filtering/sorting by price, location, and category

Why Tkinter and CustomTkinter?

- Lightweight, native to python and cross platform
- CustomTkinter modernizes UI with dark/light mode and cleaner components
- Promoted good OOP and event-driver logic
- No need for a separate backend server, we can use Python SDK or HTTP requests with supabase.

Why Reuse Supabase?

- Same backend supports both desktop and webapp
- No need to manage servers or custom DB setup
- Built-in authentication, storage, and live API's
- Allows both platforms to stay in sync with one source of truth

Agile Sprints

- ☐ Sprint 1: MVP for Desktop Application
- ☐ Sprint 2: Any possible improvements further to Desktop Application for user experience

Roles

- Team Lead/Scrum Master: Alan Sandoval
- Database Manager: Carlos Herrera
- Development Member 1: Celeste Rios
- Development Member 2: Troy Rodriguez

Group Members: Celeste Rios, Alan Sandoval, Carlos Herrera, Troy Rodriguez

Cost

- 2 Weeks

Project Architecture

- Frameworks: customtkinter, supabase, tkinter, request
- Containers: tkiner, customtkinter, import database
- Programming Languages: Python